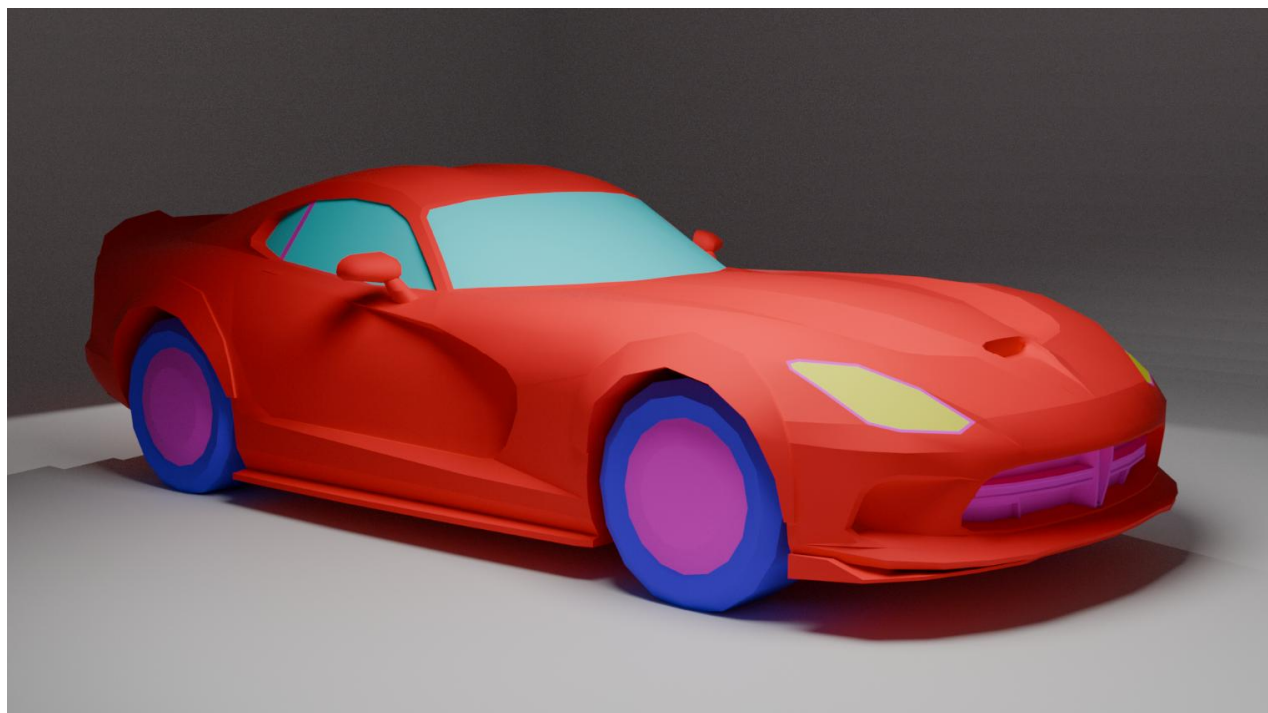


Smart Material Bridge

Blender Add-on — Documentation & User Guide

Blender 4.1+ · Adobe Substance 3D Painter



1. Overview

Smart Material Bridge is a Blender add-on that automates the full pipeline from mesh baking to final textured asset, bridging Blender and Adobe Substance 3D Painter in a single click. Handy to apply smart materials to your blender meshes. Afterwards you can adjust in Substance Painter with the baking and applying already done.

What problem does this solve?

Manually exporting low/high poly pairs, setting up bake settings in Substance Painter, applying smart materials per vertex-color ID, and re-importing textures back into Blender is a slow, error-prone process. Smart Material Bridge eliminates all manual steps.

Target Audience

This tool is aimed at 3D Artists, especially environment artists, that want an easy workflow between the baking of the mesh and applying smart materials.



2. Features

Core Features

Feature	Description
Mesh Pair Detection	Automatically finds _low / _high mesh pairs in the scene by name convention.
SP Auto-Launch & Bake	Launches Substance Painter, opens a new project with the low poly, configures bake settings, and triggers baking without interaction inside SP.
Texture Export	After baking, exports textures using the selected export preset at the chosen resolution directly to the output folder.
Texture Auto-Apply	Once SP finishes and closes, the exported textures get applied to the material within Blender.
Vertex Color Detection	Detects all unique vertex colors on the selected mesh and displays them in the panel for smart material assignment.
Smart Material Assignment	Maps vertex color IDs to specific Substance Painter Smart Materials. Each detected color can be linked to a .spsm asset; the pipeline then applies that smart material with a color-selection mask inside SP.
Export Preset Selector	Shows all the available export presets available in your SP installation.

Modular / Stretch Features

Feature	Description
Overwrite Mode	Allows re-baking into an existing output folder, safely deleting only the fbx / textures / projects sub-folders rather than the whole directory.
Named Bake Folders	Lets the artist name the output folder instead of automatically generating bake_1, bake_2, etc.
Show Vertex Colors	Show the vertex colors of the mesh to easily notice where each material should be applied.
Color Pick	Easily color pick from the detect vertex colors to modify the selections.
Overview of last bake	A complete overview of the textures that got exported, when it got exported and an option to open them in explorer.



3. Installation

Requirements

- Blender 4.1 or higher
- Adobe Substance 3D Painter
- Windows 10 / 11

Installing the Add-on

- 1 Download the ZIP**
Download Smart_Material_Bridge.zip from the provided link (or Gumroad page).
- 2 Open Blender**
Open Blender as administrator → Edit → Preferences → Add-ons.
- 3 Install**
Click 'Install...', navigate to the .zip file, and confirm.
- 4 Enable**
Find 'Smart Material Bridge' in the list and tick the checkbox to enable it.
- 5 Locate the Panel**
In the 3D Viewport, press N to open the side panel. Navigate to the 'SMB' tab.

⚠ Open Blender as administrator

This step is required. Blender needs permission to write files to the Substance Painter startup folder, which requires administrator privileges. Without running Blender as an administrator, the installation process will not be able to complete successfully.



4. Usage Guide

4.1 Preparing Your Scene

The add-on relies on a naming convention to find mesh pairs. Your objects must follow these rules:

- Your low poly mesh must end with `_low` (Barrel_low)
- Your high poly mesh must end with `_high` (Barrel_high)
- Both meshes must exist in the same scene.
- Apply vertex colors to the low poly mesh (optional)
- Select the low poly that should make use of the tool
- Reset location, rotation and scale on the objects to get the proper result

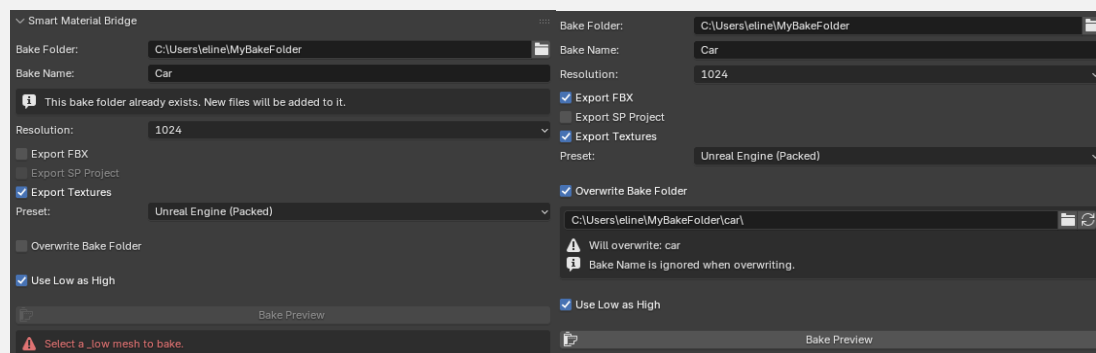
⚠ Use Clearly Distinct Vertex Colors (RGB values)



After detecting the vertex colors, verify that each color is clearly distinguishable from the others. Ideally, the colors should be spaced as far apart as possible on the color wheel.

Using highly distinct colors helps Substance Painter reliably identify each color as a separate mask, resulting in more accurate material assignments and masking.

4.2 Configuring the Panel



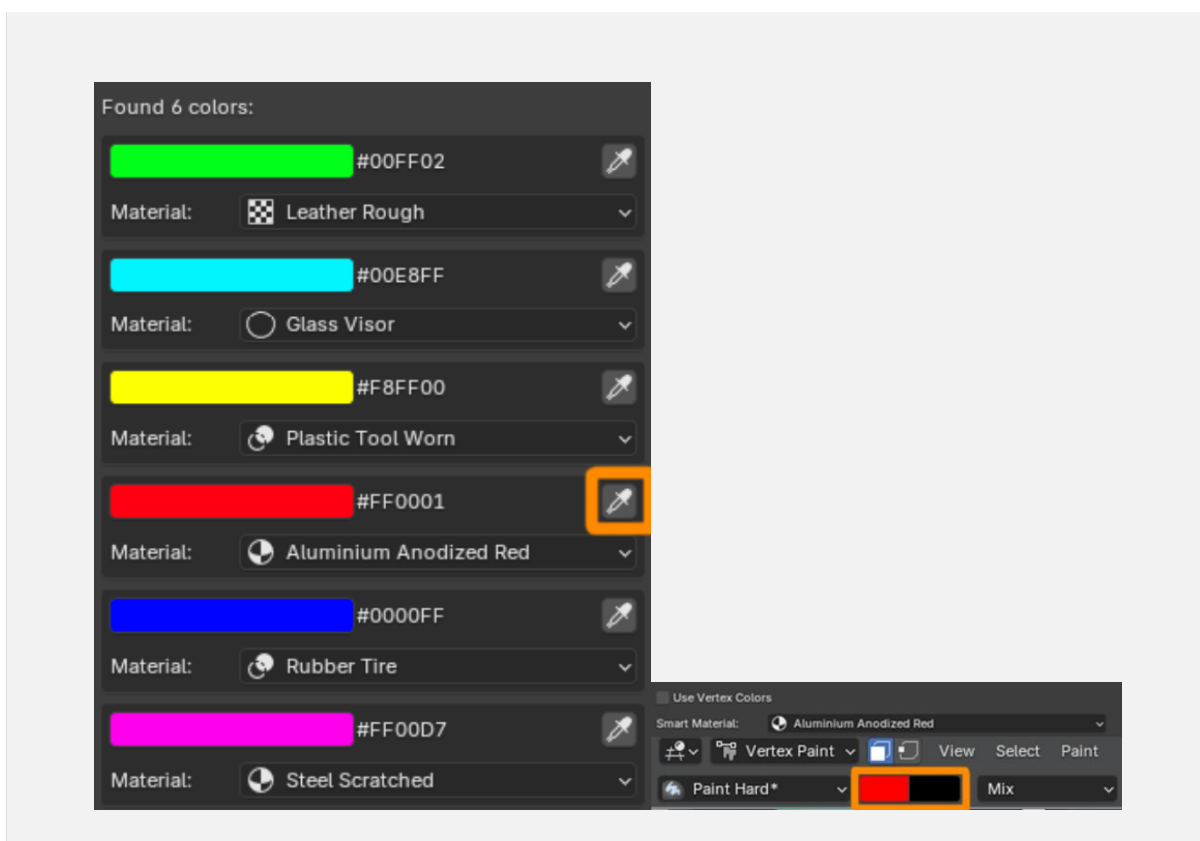
Feature	Description
Bake Folder	Root directory where all bake output folders will be created.
Bake Name	Optional name for the output subfolder. Leave blank to auto-increment (bake_1, bake_2 ...).
Resolution	Texture resolution for the bake and export. 2048 is recommended for most assets.



Feature	Description
Export FBX	Keep the exported FBX files after baking. Disable to delete them when done.
Export Blender project	Export the currently open Blender file.
Export SP Project	Save a .spp Substance Painter project file with materials applied to the projects/subfolder (requires Export FBX).
Export Textures	Export textures from SP using the selected preset.
Export Preset	Dropdown generated from your SP install. Select the export preset to use.
Overwrite Bake	Enable to write into an existing bake folder instead of creating a new one.
Use Low as High	Use the currently selected low-poly mesh as the high poly during bake in Substance Painter.

4.3 Detecting Vertex Colors

If you want to use smart material assignment, select your low poly mesh and click 'Detect Vertex Colors'. If you want to apply only one smart material to the mesh, uncheck "Use Vertex Colors".



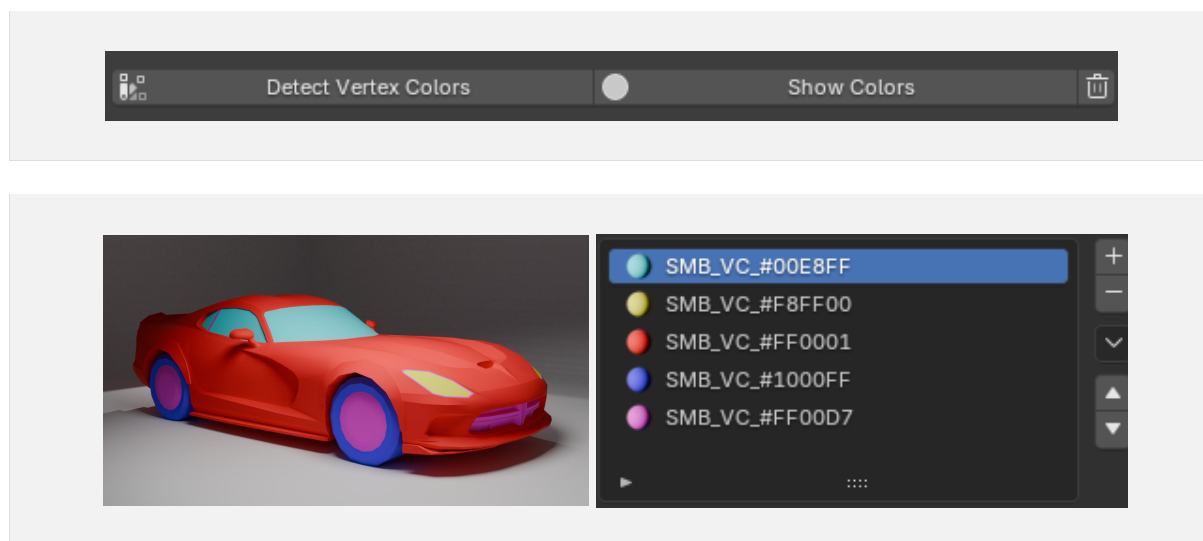
Each unique color found on the mesh appears as a row. Use the dropdown next to each color to assign a Substance Painter Smart Material. The pipeline will create a layer per color and mask it to the corresponding ID region.

Use the color picker to easily adjust faces of a certain color already specified within the panel.



4.4 Showing Vertex Colors

After detecting the vertex colors, you can choose the option “show vertex colors”. This will make multiple materials based on the amount of vertex colors applied to the mesh. The trashcan icon clears the materials from the mesh.



4.5 Running the Pipeline

- 1 Detect Vertex Colors**
Select your low poly mesh and click 'Detect Vertex Colors'. Assign smart materials to each color if desired.
- 2 Configure Settings**
Set your bake folder, resolution, export options, and preset in the SMB panel.
- 3 Click 'Bake Preview'**
The add-on exports the FBX pair, writes a config file, installs the SP bridge plugin, and launches Substance Painter automatically.
- 4 Wait — Substance Painter runs**
SP opens minimised, creates a new project, bakes Normal and ID maps, applies smart materials, exports textures, and closes itself. A progress indicator appears in the Blender panel.
- 5 Textures auto-applied**
Once SP closes and textures are detected, the Bake Watcher builds a material in Blender and connects all texture channels automatically.



4.6 Did not go as planned?

Try the following:

Problem	Solution
Crash within Substance Painter	Try baking again, configuring the settings again, restarting Blender, restarting your pc, deleting the .spp project if created.
Materials did not apply	Every export preset has different export naming conventions for the textures. This means the textures get imported into Blender, but not always automatically connected to the right input in the BSDF material node. Go to the shading tab to check and connect them manually if needed.

5. Technical overview

This section provides a brief overview of how the add-on works internally, for transparency and to demonstrate understanding of the implementation.

Blender Side

File	Description
<code>__init__.py</code>	Registers all Blender classes and scene properties when the add-on is enabled.
<code>config.py</code>	Central path configuration. Auto-detects the Substance Painter install location. All SP paths come from here — nothing is hardcoded elsewhere.
<code>handle_sp_files.py</code>	Copies the smb_bridge package into the SP startup folder before launching, and removes it again afterwards.
<code>interface/ui.py</code>	Draws the SMB panel in the N-Panel, including all conditional UI logic (warnings, error states, export options, vertex color list).
<code>interface/bake_preview.py</code>	Main operator. Validates selection, resolves the bake folder, exports the FBX pair, writes <code>bake_config.json</code> , installs SP files, launches SP, and starts the Bake Watcher.
<code>interface/bake_watcher.py</code>	Modal timer operator. Polls the texture output folder every until PNG files stop growing, then builds and applies a Principled BSDF material with all channels connected.
<code>interface/vertex_colors.py</code>	All vertex color logic in one file: reading unique colors from the mesh, storing them as scene properties, showing flat-color preview materials, and clearing them.



File	Description
interface/mesh_pairs.py	Finds _low/_high name pairs in the scene and exports each as FBX, handling visibility state save/restore for hidden objects.
interface/get_smart_materials.py	Scans SP install directories for .spsm and .spexp files to populate the smart material and export preset dropdowns.
interface/refresh_overwrite_folder.py	Sets the overwrite folder to the most recently modified subfolder inside the bake root.

Substance Painter Side

Feature	Description
smb_bridge_startup.py	SP startup script entry point. Registers start_plugin() and close_plugin() hooks.
pipeline.py	Core SP pipeline: reads config, finds FBX pairs, creates projects, triggers baking, applies smart materials, exports textures, and closes SP.
baking.py	Configures bake parameters per texture set — sets high poly mesh URL, enables Normal + ID bakers, and sets the ID source to vertex colour.
exporting.py	Resolves the export preset URL from SP's API, builds the export config, and runs the texture export.
cleanup.py	Optionally removes the FBX folder after processing and handles the finish/continue logic for multiple pairs.
load_working_dir.py	Reads bake_config.json from the SP startup directory and returns all pipeline parameters.



6. Credits & References

Feature	Description
Blender Python API	Used for operator structure, scene properties, modal timers, and node tree manipulation.
Substance Painter Python API	Adobe official SP Python API docs — Used for baking, texture export, layer stack manipulation, and event hooks.
AI Assistance	→ Claude and ChatGPT used mostly for bug fixes, reading of error codes, cleaning up of the code, generating warning and error messages.

